

Konstantinos Stamatis

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College Station, Texas, USA · Willing to Relocate and Travel Internationally

<https://kstam21.github.io/KonstantinosStamatis/about.html>

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Materials scientist and PhD candidate with 5+ years of project leadership, including 3 years developing and characterizing stimuli-responsive polymers. Polymer researcher who uses AI tools to accelerate data analysis, literature review, and experimental planning. Collaborative, adaptable, and a strong communicator, driven to solve complex problems and translate materials innovation into real business and customer value.

EDUCATION

Texas A&M University — College Station, TX

Expected Dec 2027

PhD, Materials Science & Engineering — Focus on Polymeric Actuators for a Robotic Hand and Artificial Muscles — GPA: 3.29/4.00

- Completed polymer coursework plus statistics and programming coursework covering a variety of statistical tools and techniques, applied in JMP and Python.
- Built the skills to translate research into business and customer value through Advanced Entrepreneurship and accounting coursework, from customer discovery to company launch.

Aristotle University of Thessaloniki & NCSR Demokritos — Thessaloniki & Athens, Greece

2021 - 2023

MS, Joint Program — Thesis: Development of Metal Complexes for Anticancer Drugs

- Year 1 (2021 - 2022): Chemistry Department, Aristotle University of Thessaloniki
- Year 2 (2022 - 2023): Institute of Nanoscience & Nanotechnology, NCSR Demokritos

Aristotle University of Thessaloniki — Thessaloniki, Greece

2017 - 2021

Chemistry Department, BS, Valedictorian — Major in Inorganic Chemistry

- Coursework included organic chemistry and polymer classes.

EXPERIENCE

NSF HAND Engineering Research Center (ERC) — Texas A&M University, College Station, TX

Aug 2023 - Present

Graduate Research Assistant, Dr. Taylor Ware and Dr. Svetlana Sukhishvili Laboratory, Materials Science & Engineering and Biomedical Engineering Department

- Built Python scripts through AI-assisted workflows (Claude Code) that automate DMA/DSC/TGA data analytics, including statistical analysis, automated plots, summary reports, an interactive dashboard, and publication figures, replacing manual analysis with modern digital tools and accelerating literature synthesis.
- Design, synthesize, and develop new polymer materials from concept to tested material — including recyclable Liquid Crystal Elastomers (LCEs) with Diels–Alder dynamic bonds and polydiethylsiloxane (PDES) elastomers driven by Joule-heating and electrostatic actuation — for integration into an AI-driven robotic hand, as part of the NSF HAND ERC (\$52M, 10-year initiative).
- Perform comprehensive thermomechanical characterization of actuation stress, strain, cycle life, and frequency response with high attention to detail, managing large quantities of samples, formulations, batches, and DMA/DSC/TGA data under strict protocols to analyze material properties and optimize actuation performance.
- Lead cross-laboratory collaboration with cross-functional teams and domain experts across the 5-university consortium (Texas A&M, Northwestern, CMU, MIT, FAMU), including Dr. Ryan Truby's group at Northwestern and the System Integration team, driving technology integration of actuators into the DexNex robotic hand testbed while independently solving diverse, open-ended problems across multiple parallel projects (LCE, PDES, startup testing).
- Bridge fundamental and applied research by conducting structural and thermomechanical testing and characterization of polyurethanes to address industry-relevant challenges and support product development at Bridgeway Innovations, a startup company.
- Drive commercialization efforts for advanced polymeric actuator materials, targeting application development and business and customer value in robotics and biomedical engineering.
- Plan experiments and set priorities and timelines with advisor guidance, providing regular technical updates on project milestones to advisors and the HAND ERC team and presenting LCE and PDES results and Thrust 1 milestone status to the HAND ERC industry board every 6 months.
- Mentor undergraduate researchers and serve as a Research Experiences for Undergraduates (REU) mentor, guiding two REU students (Jun 2026 - Aug 2026) in building a robotic hand prototype showcased at a HAND ERC outreach event.

NCSR Demokritos — Athens, Greece

Nov 2021 - Aug 2023

Graduate Research Assistant, Dr. Aikaterini Dendrinou-Samara, Dr. Aikaterini Raptopoulou & Dr. Vassilis Psycharis laboratory

- Led a solution-driven, 1-year project across 3 labs and partnered with cross-functional teams to synthesize and characterize four zinc complexes, evaluating their antimicrobial and anticancer properties.
- Developed novel metal complexes demonstrating in vitro antimicrobial and anticancer activities superior to those of commercially available drugs.
- Speaker at the Joint CTMNM/NAGC Conference, May 2023, Spetses, Greece.

NCSR Demokritos — Athens, Greece

Sep 2021 - Oct 2021

Research Intern, Dr. Margarit Gjoka laboratory

- Conducted research on advanced permanent magnet materials, with a focus on the synthesis and characterization of magnetic materials.
- Presented research findings at the 15th Joint MMM-INTERMAG Conference (Jan 2022, New Orleans, USA) and at the 1st Conference of NIP (Feb 2022, Tirana, Albania).

Aristotle University of Thessaloniki — Thessaloniki, Greece

Sep 2017 - Jul 2021

Undergraduate Research Assistant, Dr. Angaridis Panagiotis Laboratory

- Designed and synthesized novel copper complexes for OLED applications, conducting an in-depth study of OLED technology.

LEADERSHIP

Texas A&M University

Summer 2025 - Present

President, Society of Plastics Engineers (SPE) — global professional organization advancing knowledge and education in the plastics industry

- Plan networking events and industry site visits connecting students with polymer industry professionals, including a Mar 2026 industry seminar with Kaneka North America.
- Built the SPE Texas A&M chapter website, hand-coded in HTML/CSS/JavaScript and hosted on GitHub Pages using Claude Code.

NSF HAND Engineering Research Center (ERC) — Texas A&M University

Mar 2026 - Present

Industry Co-Chair, Student Leadership Council (SLC)

- Spearhead professional development programming connecting NSF HAND graduate students with industry leaders in robotics and advanced manufacturing.

Texas A&M University

Spring 2025 & Fall 2025

Graduate Teaching Assistant

- Teach polymer labs to 80 undergraduate students at the Baker Hughes Materials Lab, Texas A&M University. - Supervise and mentor laboratory students in preparing detailed 30-page reports, executing advanced characterization and assay development protocols, and managing \$200K worth of lab equipment and materials.	Fall 2023 - Summer 2025
Texas A&M University <i>Vice President, Materials Science & Engineering Graduate Student Advisory Council</i> Collaborated with Materials Science and Engineering department leadership to enhance student life and expand academic resources.	Fall 2023 - Present
Texas A&M University <i>Founder & President, Hellenic Student Association</i> - Founded, led, and secured official recognition for the organization at Texas A&M University, inspiring student engagement, recruiting faculty support, and fostering active participation. - Collaborated with Halliburton Engineering Global Programs leadership and Greek faculty to expand Texas A&M's international student recruitment in Greece and Cyprus.	Spring 2025 - Present
Organizations - Member, Material Advantage at Texas A&M University - Member, Minerals, Metals & Materials Society (TMS) - Member, American Society for Metals (ASM) - Member, American Ceramic Society (ACerS) - Member, Association for Iron & Steel Technology (AIST)	Oct 2018
Erasmus+ — Berlin, Germany <i>Greek Representative, European Youth Exchange Program</i> - Selected and fully funded by the United Societies of the Balkans, a non-profit organization (NGO). - Promoted youth mobility and collaborated with teams from five different countries: Bulgaria, Poland, Spain, Romania, and Italy. - Addressed environmental awareness issues in European countries and explored renewable energy sources.	Sep 2018 - Aug 2019
Aristotle University of Thessaloniki <i>Social Manager, Chemistry Professional Organization "ReAcTION"</i> Organized events and outreach programs to promote the field of chemistry to the public.	
PUBLICATIONS	
Enhanced Actuation Stability of Liquid Crystal Elastomer Actuators With Diels–Alder Dynamic Bonds. Stamatis, K. et al. Small 2026, e74705. https://doi.org/10.1002/sml.74705 (Impact Factor 12.1)	
Engineered Living Materials Suppress Uropathogenic E. coli Growth and Invasion of Urothelial Cells Through Sustained Probiotic Release. Kalairaj, M. S., ..., Stamatis, K., ..., Ware, T. H. bioRxiv 2026 (Preprint). https://doi.org/10.64898/2026.09.15.751868	
Synthesis, Structural and Spectroscopic Characterization, and DNA Binding Studies of Chiral Zn(II) Complexes with Enantiopure Schiff Base Ligands. Stamatis, K. et al. Manuscript in preparation, 2026.	
ACHIEVEMENTS	
SPE International Polyolefins Conference Scholarship Competitive graduate-level scholarship awarded for outstanding research excellence at Texas A&M University.	Feb 2026
Texas Soft Matter, Rice University Presentation Award Winner selected from 43 student participants for excellence in research presentation.	Aug 2025
Gerondelis Foundation Grant Awarded scholarship supporting Greek nationals in U.S. graduate programs.	Jul 2025
Henry Kahn Memorial Scholarship <i>Society of Plastics Engineers (SPE) South Texas Section</i> Awarded for excellence in academics and research among graduate and undergraduate students across Texas.	Apr 2025
Material Advantage Congressional Visits Day (CVD) Participant <i>Participated in the 2025 Material Advantage Congressional Visits Day in Washington, D.C.</i> - Awarded a \$320 travel grant to support participation in the event. - Engaged with Texas Senators and Representatives to advocate for the support of education and research in the physical sciences and engineering, emphasizing their role in driving innovation and scientific progress.	Apr 2025
Aggies Invent: Problems Worth Solving Competition, Texas A&M Entrepreneurship Department <i>First Place Award for Designing the Most Innovative Prototype</i> Applied ideation and rapid prototyping to address costly and life-threatening catheter-associated UTIs.	Nov 2024
SKILLS	
Material characterization: Dynamic Mechanical Analysis (DMA), Universal Testing Machine (UTM), Differential Scanning Calorimetry (DSC), Gel Permeation Chromatography (GPC), Infrared (IR) Spectroscopy, Scanning Electron Microscopy (SEM), Thermogravimetric Analysis (TGA), Polarized Optical Microscopy (POM), Powder X-ray Diffraction (XRD), Wide-Angle X-ray Scattering (WAXS), Vibrating Sample Magnetometry (VSM), Fluorescence Spectroscopy, H Nuclear Magnetic Resonance (NMR), UV/Vis Spectroscopy, Circular Dichroism (CD)	
Computer science and data science: Python (AI-assisted via Claude Code), JMP (statistical analysis), Advanced Excel, MATLAB, Git/GitHub, automated data visualization and interactive dashboards, AI-assisted workflows (Claude, Claude Code, Open AI Codex)	
Web development: HTML/CSS/JavaScript and GitHub Pages; built a personal research website and the SPE Texas A&M chapter website	
Safety: Texas A&M EHS laboratory safety, chemical and hazardous waste, and equipment-specific training	
Materials and processing: Polymer synthesis, reprocessing and recycling, 3D printing of LCEs, fiber fabrication	
Lab and design software: SolidWorks, Origin, TRIOS, ImageJ, ChemDraw, Mercury, Microsoft Office	
Languages: Greek (Native), English (Fluent)	
Sports: Competitive level in soccer	
Music: Piano, trumpet, guitar, and Byzantine music	